week07-1.在電腦用程式控制聲音的輸入

```
// week07_1_processing_audio_???
// File-Preference
// 想要在電腦用程式控制聲音的輸入
// 方法1:Processing的Sound 外掛，方法2:Minim的外掛
// Sketch - Library - Manage Library
// 或者用 google:processing sound
// Moodle【上課用軟體】有一份sound.zip
// 把三個資料夾的Sound 放在「文件-Processing-Library」目錄裡
// 可以直接把檔案拉進程式

import processing.sound.*;
SoundFile sound;
void setup(){
  size(300, 300);
  sound = new SoundFile(this, "music.mp3"); //檔名
  sound.play(); //準備好 music.mp3 拉進程式，撥放
}
void draw(){

}
```

```
// week07_1_processing_audio_Soundfile_play
// File-Preference
// 想要在電腦用程式控制聲音的輸入
// 方法1:Processing的Sound 外掛， 方法2:Minim的外掛
// Sketch - Library - Manage Library
// 或者用 google:processing sound
// Moodle【上課用軟體】有一份sound.zip
// 把三個資料夾的Sound 放在「文件-Processing-Library」目錄裡

import processing.sound.*;
SoundFile sound;
void setup(){
  size(500, 500);
  sound = new SoundFile(this, "RASEN Acoustic Ver - 9Lana.mp3");//檔名
  sound.play(); //準備好 music.mp3 拉進程式，撥放
}
void draw(){
   background(0);
   translate(250, 250);
   //translate(mouseX, mouseY);
   //println(mouseX, mouseY);
   rotate( radians(frameCount));
   rect(-200, -200, 400, 400);

}
```

/week07_2_processing_audio_two_SoundFile

```
// week07_2_processing_audio_two_SoundFile
// 修改自 week07_1_processing_audio_Soundfile_play

import processing.sound.*;
SoundFile sound;
SoundFile sound2;

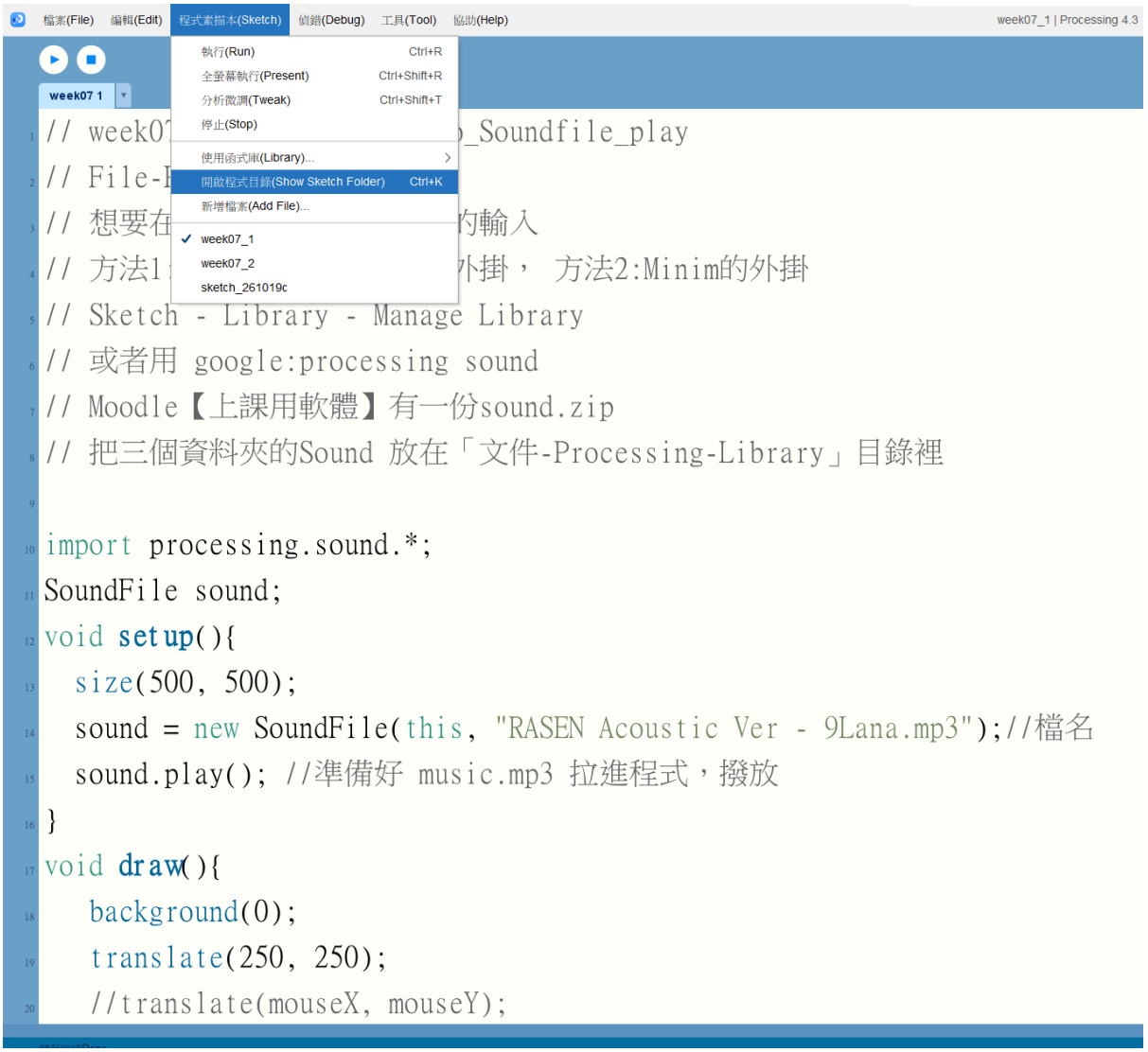
void setup(){
  size(500, 500);
  sound = new SoundFile(this, "music.mp3");//檔名
  sound2 = new SoundFile(this, "mi.wav");
  sound.play(); //準備好 music.mp3 拉進程式，撥放
}
void draw(){
   background(0);
   translate(250, 250);
   //translate(mouseX, mouseY);
   //println(mouseX, mouseY);
   rotate( radians(frameCount));
   rect(-200, -200, 400, 400);
}

void mousePressed(){
   sound2.play();
}
```

```
void setup(){
  size(500, 500);
  sound = new SoundFile(this, "music.mp3");//檔名
  sound2 = new SoundFile(this, "mi.wav");
  sound.play(); //準備好 music.mp3 拉進程式，撥放
}
void draw(){
   background(0);
   translate(250, 250);
   //translate(mouseX, mouseY);
   //println(mouseX, mouseY);
   rotate( radians(frameCount));
   rect(-200, -200, 400, 400);
}

void mousePressed(){
   sound2.play();
}
```

//week07_A_查看程式檔案



//week07_3_processing_audio_doremi

```
//week07_3_processing_audio_doremi
import processing.sound.*;
SoundFile Do, Re, Mi;
void setup(){
  size(300,200);

  Do = new SoundFile(this, "do.wav");
  Re = new SoundFile(this, "re.wav");
```

```
   Mi = new SoundFile(this, "mi.wav");

}
void draw(){
    rect(0, 0, 100, 200);
    rect(100, 0, 100, 200);
    rect(200, 0, 100, 200);
}

void mousePressed(){
  if(mouseX < 100){
     Do.play();
  } else if(mouseX < 200){
     Re.play();
  } else if(mouseX < 300){
     Mi.play();
  }

}
```

```
//week07_3_processing_audio_doremi
import processing.sound.*;
SoundFile Do, Re, Mi;
void setup(){
    size(300,200);

    Do = new SoundFile(this, "do.wav");
    Re = new SoundFile(this, "re.wav");
    Mi = new SoundFile(this, "mi.wav");

}
void draw(){
     rect(0, 0, 100, 200);
     rect(100, 0, 100, 200);
     rect(200, 0, 100, 200);
}

void mousePressed(){
  if(mouseX < 100){
     Do.play();
  } else if(mouseX < 200){
     Re.play();
  } else if(mouseX < 300){
     Mi.play();
  }

}
```

//week07_4_processing_serial_audio_doremi

```
//week07_4_processing_serial_audio_doremi
//修改自week07_3_processing_audio_doremi
//修改自week05_3_processing_audio_doremi

import processing.serial.*;
Serial myPort;
import processing.sound.*;
SoundFile Do, Re, Mi;
void setup(){
  size(300,200);
  myPort = new Serial(this,"COM4", 9600); //要加這行
  //將do re mi 拉進程式
  Do = new SoundFile(this, "do.wav");
  Re = new SoundFile(this, "re.wav");
  Mi = new SoundFile(this, "mi.wav");

}
void draw(){
   rect(0, 0, 100, 200);
   rect(100, 0, 100, 200);
   rect(200, 0, 100, 200);
}

void mousePressed(){
 if(mouseX < 100){
    Do.play();
    myPort.write('1');
 } else if(mouseX < 200){
    Re.play();
    myPort.write('2');
 } else if(mouseX < 300){
    Mi.play();
    myPort.write('3');
 }
}
void mouseReleased(){
 myPort.write('0');

}
```

```
//week07_4_processing_serial_audio_doremi
//修改自week07_3_processing_audio_doremi
//修改自week05_3_processing_audio_doremi
import processing.serial.*;
Serial myPort;
import processing.sound.*;
SoundFile Do, Re, Mi;
void setup(){
   size(300,200);
   myPort = new Serial(this,"COM4", 9600); //要加這行
   //將do re mi 拉進程式
   Do = new SoundFile(this, "do.wav");
   Re = new SoundFile(this, "re.wav");
   Mi = new SoundFile(this, "mi.wav");
}
void draw(){
    rect(0, 0, 100, 200);
    rect(100, 0, 100, 200);
    rect(200, 0, 100, 200);
}
void mousePressed(){
  if(mouseX < 100){
     Do.play();
     myPort.write('1');
  } else if(mouseX < 200){
     Re.play();
     myPort.write('2');
  } else if(mouseX < 300){
     Mi.play();
     myPort.write('3');
  }
}
void mouseReleased(){
  myPort.write('0');
}
```

// week07-4-arduino-processing

```
//week07_4_processing_serial_audio_doremi


// week07-4-arduino-processing
// 修改自 week05-3-arduino-do-re-mi

void setup() {
  pinMode(8, OUTPUT); //Buzzer 8 聲音
  pinMode(10, OUTPUT); //對應'0'
  pinMode(11, OUTPUT); //對應'1'
  pinMode(12, OUTPUT); //對應'2'
  pinMode(13, OUTPUT); //對應'3'


  Serial.begin(9600);//USB Serial 開始傳輸 速度9600bps
  tone(8, 523, 100); delay(200); //Do
  tone(8, 587, 100); delay(200); //Re
  tone(8, 659, 100); delay(200); //Mi
  tone(8, 587, 100); delay(200); //Re
  tone(8, 523, 100); delay(200); //Do

}

char c = '0'; //在外宣告變數
void loop() {
  // put your main code here, to run repeatedly:
    if(Serial.available()){
      c = Serial.read();
    }


    for(int i=10; i<=13; i++) digitalWrite(i, LOW);
    if(c>='0' && c<='3') digitalWrite(c-'0'+10,HIGH);
    if (c=='0') noTone(8);// 不發出聲音
    if (c=='1') tone(8,523);// Do 1秒
    if (c=='2') tone(8,587);// Re 1秒
    if (c == '3') tone(8, 659); // 3 (Mi)  659 Hz, 1秒
    if (c == '4') tone(8, 698); // 4 (Fa)  698 Hz, 1秒
    if (c == '5') tone(8, 784); // 5 (Sol) 784 Hz, 1秒
    if (c == '6') tone(8, 880); // 6 (La)  880 Hz, 1秒
  }
```